Loan Repayment Prediction

A Full Machine Learning Pipeline for Credit Risk Assessment

Python | Scikit-Learn | TensorFlow | SMOTE | Gradio

Edwin Korir · 2026

Agenda

01 **Problem Overview**
Why loan repayment prediction matters

02 **Dataset & Features**
614 applications, 12 features, target variable

03 **Data Preprocessing**
Cleaning, encoding, SMOTE, scaling

04 **Exploratory Analysis**
Key insights and distributions

05 **Model Training**
6 classifiers: LR, KNN, SVM, RF, DT, ANN

06 **Results & Comparison**
Accuracy, ROC-AUC, model benchmarks

07 **Conclusions**
Findings, recommendations, next steps

Problem Overview

Why does loan repayment prediction matter?

Credit Risk

Lenders face significant losses when borrowers default. Data-driven risk models reduce this exposure.

Faster Decisions

Manual loan reviews are slow and inconsistent. ML models deliver instant, standardised decisions.

Fairness

Objective models reduce human bias in credit decisions, promoting equitable access to finance.

Our Goal

Predict Loan_Status (Y/N) using 12 applicant features — accuracy, precision, and recall all matter.

Dataset & Features

614
Loan Applications

12
Input Features

68%
Approved (raw)

32%
Rejected (raw)

Feature	Type	Description
Gender / Married	Categorical	Demographic attributes
Dependents	Ordinal	0, 1, 2, 3+
Education / Self_Employed	Categorical	Graduate? Self-employed?
ApplicantIncome / CoapplicantIncome	Numeric	Monthly income (right-skewed)
LoanAmount / Loan_Amount_Term	Numeric	Loan size & repayment term
Credit_History	Binary	1 = meets guidelines (strongest predictor)
Property_Area	Categorical	Rural / Semiurban / Urban
Loan_Status ← TARGET	Binary	Y = Repaid, N = Not Repaid

Data Preprocessing Pipeline

1	2	3	4	5
Missing Values	**Encoding**	**Transformation**	**SMOTE**	**Scaling**
Mode imputation for categoricals Median imputation for LoanAmount	Label encoding for: Gender, Married, Education, Self_Employed, Property_Area	Square root on ApplicantIncome & CoapplicantIncome to reduce skew	Synthetic oversampling to balance 68/32 → 50/50 class split	MinMaxScaler → normalise all features to range [0, 1]

Exploratory Data Analysis — Key Findings

78%

of applicants are graduates

Higher education correlates with better repayment ability

85%

credit history → approval

Credit_History is the single strongest predictor of Loan_Status

Semiurban

highest repayment area

Property area is a meaningful geographic predictor

Right-skewed

income distributions

Sqrt transformation applied to ApplicantIncome & CoapplicantIncome

~13.6%

avg missing per column

Mode/median imputation used; no columns dropped

500 : 214

male vs other applicants

Dataset skews male — fairness auditing recommended

Models Trained

Logistic Regression
Baseline

Solver: saga, max_iter: 500
Baseline classifier

K-Nearest Neighbors
Deployed

k tuned over range 1–20
Selected for Gradio deployment

Support Vector Machine
Classical

RBF kernel
Competitive precision

Random Forest
★ Best

1000 estimators
Tuned leaf nodes (2–24)

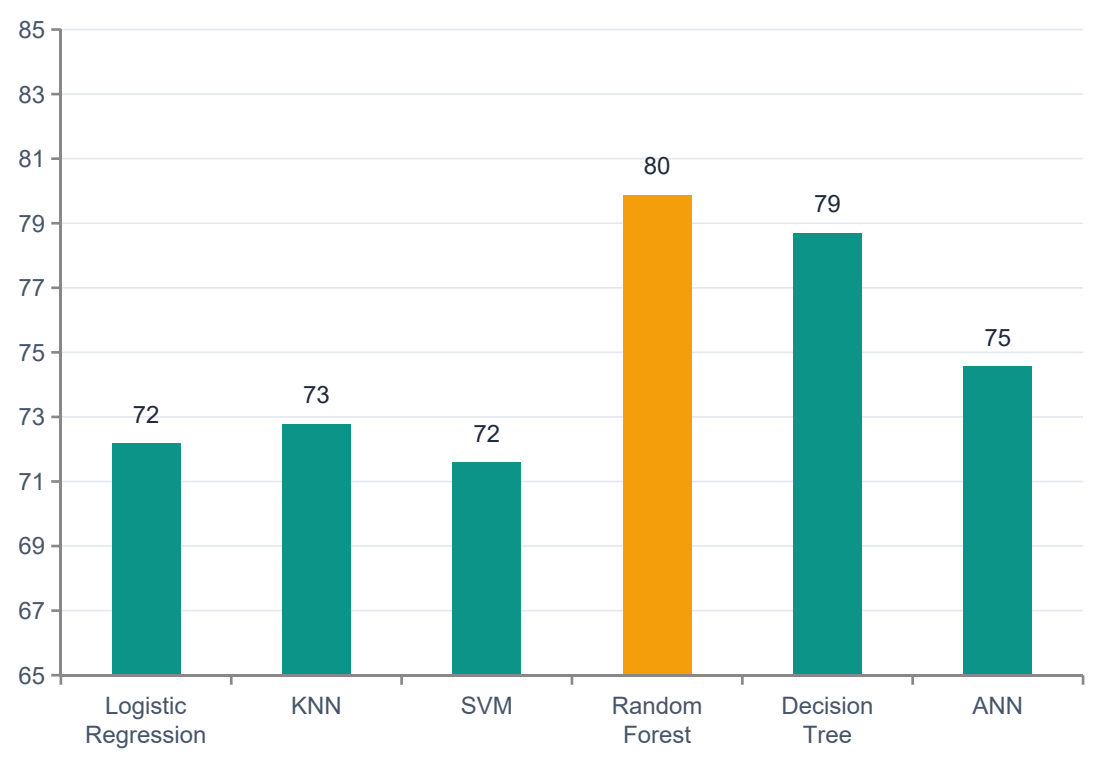
Decision Tree
Classical

Max depth tuned
Highly interpretable

Neural Network (ANN)
Deep

10 epochs, Adam optimizer
Binary Crossentropy loss

Model Performance Comparison



Top Model

Random Forest

Accuracy	79.88%
Precision	~83.7%

1000 estimators, tuned leaf nodes, robust to outliers

⚡ KNN selected for Gradio deployment (simplicity + speed) | All models trained on SMOTE-balanced data

Conclusions & Recommendations

Key Conclusions

- Credit History is the strongest single predictor
- SMOTE was critical to address 68/32 class imbalance
- Income skew required sqrt transformation
- Random Forest outperforms all single classifiers
- Semiurban area shows highest repayment rates

Recommendations

- Use Random Forest for production deployment
- Collect more diverse demographic data
- Add features: DTI ratio, employment tenure
- Retrain periodically to prevent model drift
- Conduct fairness audit before go-live

Thank You

Loan Repayment Prediction — A Machine Learning Approach

Email

ekorir99@gmail.com

GitHub

github.com/Edwinkorir38

LinkedIn

linkedin.com/in/edwin-korir-90a794382

Try the Live Demo on Streamlit